MILITARY SPECIFICATION

PRIMER, PERCUSSION, ELECTRIC, M75 LOADING, ASSEMBLING, AND PACKING (INCLUDING LOADING OF PERCUSSION-ELECTRIC IGNITION ELEMENT T21E1).

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of primer designated as Primer, Percussion, Electric, M75.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

JAN-P-223 - Powder, Black.
MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.

DRAWINGS

ORDNANCE CORPS

7549264 - Primer, Percussion-Electric, M75; Loading and Assembly.
8796535 - Box, Set-Up, Packing, Ammunition for Primers, Percussion, Electric, T106 or Electric T107. Assembly Details and Packing, For Domestic Use Only.

FSC 1390

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8796536 - Box, Packing, For Primer, Percussion, Electric, T106 or Electric, T107, Assembly, Details and Packing. For Domestic Use Only.
8796678 - Box, Set-Box, Set-Up, Packing, Ammunition for Primers Percussion, Electric, T106 or Electric T107. Assembly, Details, Packing and Marking. For Overseas Use Only.
8796679 - Box, Packing, Ammunition, for Primers, Percussion, Electric, T106 and Electric, T107. Assembly, Details and Packing. For Overseas Use Only.

PUBLICATIONS

ORDNANCE INSPECTION HANDBOOK

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
GL-8801948 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Primers.- The primers shall comply with all requirements specified on Drawing (dwg.) 7549264 and with all requirements specified in applicable specifications.

3.3 Moisture content of black powder.-The moisture content of the black powder at the loading station, at the time of loading, shall not exceed 0.5 percent when determined as specified in 4.3.1.

3.4 Moisture content of ignition charge composition.-The moisture content of the ignition charge composition at the loading station at the time of loading the ignition element shall not exceed 0.1 percent when determined as specified in 4.3.2.

3.5 Electrical resistance.-The primers shall have an electrical resistance of not less than 0.80 ohms nor more than 1.80 ohms and shall not function when tested as specified in paragraph 4.3.3.

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3.6 Non-functioning.

3.6.1 Percussion phase.-The primer shall not function under an applied energy value of 47.64 inch ounces minimum (min.) when tested as specified in 4.3.4.1.

3.6.2 Electrical phase.-The primers shall not function when tested as specified in 4.3.4.2.

3.7 Functioning.

3.7.1 Percussion phase.-The primers shall function under an applied energy value of 158.80 inch ounces maximum (max.) when tested as specified in 4.3.5.1.

3.7.2 Electrical phase.-The primers shall function when tested as specified in 4.3.5.2.

3.8 Firing.-(Proving ground).

3.8.1 Percussion.-The primers shall function when tested as specified in 4.3.6.1.

3.8.2 Electrical.-The primers shall function when tested as specified in 4.3.6.2.

3.9 Workmanship.-All parts shall be dry and free of burrs, chips, dirt, grease, rust and foreign matter prior to, and following the loading and assembling operations. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services, acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

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4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet his approval, prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (gages, measuring, and testing equipment), and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.2 and 4.3 (except proving ground tests), and the other documents referenced herein. The contractor shall not be restricted to the inspection station nor the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analyses on all material procured directly by the contractor.

f. Number of items in the lot.

g. Date submitted.

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4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied and (b) Government product inspection. The Government inspector shall also ascertain prior to acceptance that the product complies with all the quality assurance provisions of other specifications referenced in any of the contractual documents.

4.1.3.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.3.2 Product inspection.-Physical inspection by the Government of contractor accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records, Specification MIL-G-2550 will apply.

4.1.3.3 Deviations and deficiencies.-Deviations from the prescribed or agreed upon procedures, or instances of poor practice which might have an effect on the quality of the product, shall be called to the attention of the contractor immediately. To avoid interference, with operations, the contractor shall designate a responsible official, or officials, to whom the Government inspector will report such instances. Failure of the contractor to correct deficiencies promptly shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

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4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of primers produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot number. Each lot shall contain:

a. Ignition element components of one interfix lot number from one manufacturer.

b. Percussion primers of one interfix lot number from one manufacturer.

c. Primer metal components of one interfix lot number from one manufacturer.

d. Black powder of one interfix lot number from one manufacturer.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Ignition cup assembly, prior to loading (see dwg. 8830967 covering a detail of dwg. 7549264).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Bridge wire missing or broken	Visual
Minor:	AQL 0.40 percent	
201.	Evidence of poor workmanship (see 3.9)	Visual

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4.2.2.2 Percussion-electric ignition element T21E1, prior to assembly of plunger (see dwg. 8830966 covering a detail of dwg. 7549264).

Categories	Defects	Method of inspection
Critical:		
1.	Depth to top of primer, min.	Gage
Major:	0.25 percent	
101.	Primer missing or damaged	Visual
102.	Seal incomplete at joint between primer and cup sidewall	Visual
Minor:	AQL 0.40 percent	
201.	Evidence of poor workmanship (see 3.9)	Visual

4.2.2.3 Percussion-electric ignition element T21E1 (see dwg. 8830966 covering a detail of dwg. 7549264).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Ignition charge above flush	Visual
102.	Insulation punctured, cut or torn	Visual
103.	Ignition charge on exterior surface	Visual
Minor:	AQL 1.00 percent	
201.	Total length, max.	Gage
202.	Evidence of poor workmanship (see 3.9)	Visual

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4.2.2.4 Container, charge assembly (see dwg. 7549263 covering a detail of dwg. 7549264.)

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Weight of charge, min.	Balance
102.	Closure assembly above flush	Visual-Manual
103.	Assembly damaged	Visual
Minor:	AQL 1.00 percent	
201.	Closure assembly incompletely sealed	Visual
202.	Evidence of poor workmanship (see 3.9)	Visual

4.2.2.5 Assembly, prior to inserting ignition element (see dwg. 7549264).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Body damaged	Visual
Minor:	AQL 0.40 percent	
201.	Foreign matter on interior surface (see 3.9)	Visual

4.2.2.6 Assembly, prior to assembling container charge assembly (see dwg. 7549264).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Closing disc missing	Visual
Minor:	AQL 1.00 percent	
201.	Closing disc punctured	Visual
202.	Foreign matter in container charge cavity (see 3.9)	Visual

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4.2.2.7 Assembly (see dwg. 7549264).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Body and container joint seal incomplete	Visual
102.	Plunger or plunger insulator above flush	Visual
103.	Container cup cracked or punctured	Visual
104.	Container charge assembly not fully seated	Manual
Minor:	AQL 2.50 percent	
201.	Total length, max.	Gage
202.	Diameter of flange, max.	Gage
203.	Coating missing on container assembly	Visual
204.	Evidence of poor workmanship (see 3.9)	Visual
205.	Marking misleading or unidentifiable	Visual

4.2.2.8 Box, set-up, packing, domestic (see dwg. 8796535).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Assembly damaged so that contents are exposed or liable to become exposed	Visual
102.	Sealing strip incomplete, badly wrinkled, or not in contact with surface of box for at least 1/2 inch around entire body	Visual
Minor:	AQL 0.40 percent	
201.	Contents move upon shaking	Manual

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4.2.2.9 Box, packing, domestic (see dwg. 8796536).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Strapping missing, broken or loose....	Visual-Manual
102.	Board broken or split	Visual
103.	Contents exposed	Visual
Minor:	AQL 1.50 percent	
201.	Strapping improperly assembled	Visual-Manual
202.	Contents move when shaken	Manual
203.	Marking misleading or unidentifiable .	Visual

4.2.2.10 Box, set-up, packing, overseas (see dwg. 8796678).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Assembly damaged so that contents are exposed or liable to become exposed	Visual
102.	Sealing strip incomplete, badly wrinkled, or not in contact with surface of box for at least 1/2 inch around entire box	Visual
Minor:	AQL 1.50 percent	
201.	Contents move upon shaking	Manual
202.	Label missing (when applicable)	Visual
203.	Marking misleading or unidentifiable.	Visual

4.2.2.11 Box, packing, overseas (see dwg. 8796679).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Hardware or strapping missing, broken or loose	Visual-Manual
102.	Board broken or split	Visual
103.	Contents exposed	Visual

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Minor:	AQL 2.50 percent	
201.	Hardware or strapping improperly assembled	Visual-Manual
202.	Car seal missing, unsealed or improperly positioned	Visual-Manual
203.	Handle missing or insecure	Visual-Manual
204.	Contents move when shaken	Manual
205.	Marking misleading or unidentifiable	Visual

4.2.3 Testing.-The test equipment shall be in accordance with 4.2.4.

4.2.3.1 Initial production.-The first lot produced in accordance with 4.2.1, and all successive lots, prior to Government acceptance of the initial production quantity, shall be inspected as required by this specification, except that the sampling plans for non-functioning, functioning, firing and electrical resistance tests shall be as specified below. The contractor shall continue this plan of inspection until three consecutive lots have complied with the requirements.

4.2.3.1.1 Non-functioning (see 3.6).

4.2.3.1.1.1 Percussion phase (see 3.6.1) - Critical defect.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, Table IV A, code letter L, and an AQL of 0.065 percent. The test shall be performed as specified in 4.3.4.1.

4.2.3.1.1.2 Electrical phase (see 3.6.2) - Major defect.-The sampling plan for this test shall be as specified in 4.2.3.1.1.1. The test shall be performed as specified in 4.3.4.2. *

4.2.3.1.2 Functioning (see 3.7).

4.2.3.1.2.1 Percussion phase (see 3.7.1) - Major defect.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, code letter O and an AQL of 0.15 percent. The test shall be performed as specified in 4.3.5.1.

4.2.3.1.2.2 Electrical phase (see 3.7.2) - Major defect.-The sampling plan for this test shall be as specified in 4.2.3.1.2.1. The test shall be performed as specified in 4.3.5.2.

4.2.3.1.3 Firing (see 3.8).

4.2.3.1.3.1 Percussion (see 3.8.1) - Major defect.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, Table IV A, code letter G and an AQL of 0.65 percent. The test shall be performed as specified in 4.3.6.1.

* Primers tested for the non-functioning electrical phase shall not be returned to the lot.

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4.2.3.1.3.2 Electrical (see 3.8.2) - Major defect.-The sampling plan for this test shall be as specified in 4.2.3.1.3.1. The test shall be performed as specified in 4.3.6.2.

4.2.3.2 Normal production.-After acceptance of the initial production lots, testing shall be as follows:

4.2.3.2.1 Moisture content at loading station at time of loading.-The contractor shall provide adequate controls to assure the compliance of each lot with the applicable requirements. The contractor shall test for verification at least one completed primer assembly from each eight hours production. Composite samples shall not be used in determining compliance. The following composition and powder shall be tested as specified and any failure to comply with the applicable requirement shall be cause for rejection of the eight hours production represented by the sample.

4.2.3.2.1.1 Black powder (see 3.3) - Major defect.-The moisture content shall be determined as specified in 4.3.1.

4.2.3.2.1.2 Ignition charge composition (see 3.4) - Major defect.-Testing shall be performed as specified in 4.3.2.

4.2.3.2.2 Electrical resistance (see 3.5) - Critical defect.-This test shall be conducted 100 percent. Any assembly which fails to comply with the applicable requirements shall be classed defective and removed from the lot. The test shall be performed as specified in 4.3.3.

4.2.3.2.3 Non-functioning (see 3.6).

4.2.3.2.3.1 Percussion phase (see 3.6.1) - Critical defect.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, using Table IV-A, code letter I and an AQL of 0.25 percent. The test shall be performed as specified in 4.3.4.1.

4.2.3.2.3.2 Electrical phase (see 3.6.2) - Critical defect.-The sampling plan for this test shall be as specified in 4.2.3.2.3.1. The test shall be performed as specified in 4.3.4.2. *

4.2.3.2.4 Functioning (see 3.7).

4.2.3.2.4.1 Percussion phase (see 3.7.1) - Major defect.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, using code letter M and an AQL of 0.15 percent. The test shall be performed as specified in 4.3.5.1.

4.2.3.2.4.2 Electrical phase (see 3.7.2) - Major defect.-The sampling plan for this test shall be as specified in 4.2.3.2.4.2. The test shall be performed as specified in 4.3.5.2.

* Primers tested for the non-functioning electrical phase shall not be returned to the lot.

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4.2.3.2.5 Firing (see 3.8).

4.2.3.2.5.1 Percussion (see 3.8.1) - Major defect.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, using Table IV-A, code letter E and an AQL of 1.5 percent. The test shall be performed as specified in 4.3.6.1.

4.2.3.2.5.2 Electrical (see 3.8.2) - Major defect.-The sampling plan for this test shall be as specified in 4.2.3.2.5.1. The test shall be performed as specified in 4.3.6.2.

4.2.3.2.6 Determination of ignition charge composition (see dwg. 8830966) - Major defect.-A sufficient amount of ignition charge composition, representative of each 8 hours production of primers, shall be selected for analysis. The analysis shall be performed as specified in 4.3.7.

4.2.3.2.7 Profile and alignment (see dwg. 7549264) - Major defect.-The primers shall be gaged 100 percent for profile and alignment. Any primer which fails to meet the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4.

4.2.3.2.8 Check test of percussion elements.-If the total elapsed time between the original acceptance of any percussion element lot and the assembly of that lot into a complete artillery primer lot exceeds two years, or if the elements have been subjected to any adverse conditions, however brief, at any time since previous tests, the element lot shall be subjected to, and must satisfactorily pass the check test specified in the applicable percussion element specification, immediately before the elements are assembled into the artillery primers. This test shall be performed by the contractor on elements selected by the Government inspector at the facility assembling the elements into the artillery primers, and shall be conducted at Government expense without cost to the contractor (see 6.2) who loaded the elements or the contractor assembling the elements into the artillery primers and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc. permitted, while the element lot was under the jurisdiction of either contractor.

4.2.4 Inspection equipment.-For the performance of the examinations and tests specified in 4.2 and 4.3, when applicable, the contractor shall have available, and utilize correctly, all necessary gages, measuring, and testing equipment, identified by Index of Inspection Equipment Lists No. GL-8801948. When inspecting the contractor's inspection equipment the Government inspector will determine that the contractor has available, and utilizes correctly,

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gages, measuring and testing equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. The contractor will have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards, and instruments will be periodically checked by authorized Government personnel.

4.3 Test methods and procedures.

4.3.1 Determination of moisture content of black powder.-The moisture content of the black powder shall be determined in accordance with the procedure specified in Specification JAN-P-223.

4.3.2 Determination of moisture content of ignition charge composition.

4.3.2.1 Preparation of acetyl-chloride solution.-A 118 milliliter (ml.) portion of acetyl chloride, contained in a liter volumetric flask, shall be dissolved in toluene and then diluted with toluene to the mark. A 10 ml. portion of pyridine shall be added to the toluene solution to stabilize the acetyl chloride.

4.3.2.2 Procedure.-An accurately weighed one gram portion of the sample shall be placed in a glass-stoppered weighing bottle (outside ground) of approximately 100 ml. capacity. A 20 ml. portion of anhydrous methanol shall be added by means of a pipette. The bottle shall be stoppered and the mixture agitated briefly. The mixture shall be allowed to stand for one hour to insure complete absorption of the moisture by the methanol. A 10 ml. portion of the prepared acetyl chloride solution shall be transferred by means of a pipette to a 250 ml. glass-stoppered Erlenmeyer flask. The stoppered flask and contents shall be placed in an ice bath for two minutes. A two ml. portion of pyridine shall be added and the mixture shaken briefly. A 10 ml. portion of the previously prepared methanol sample shall be transferred by means of a pipette to the flask. The mixture shall be agitated again and then allowed to stand for two minutes. A one ml. portion of absolute ethanol shall be added, and after 5 minutes, an additional 25 ml. portion of ethanol shall be added. After 10 minutes the solution shall be titrated with 0.4 Normal sodium hydroxide solution using a one percent phenolphthalein solution as an indicator. A blank determination shall be run on a solution containing all reagents. The amount of moisture is indicated by the increase in acidity of

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the sample over the blank. The percentage of moisture shall be calculated as follows:

$$\text{Percent moisture} = \frac{0.9\ (V-v)\ N}{W}$$

where:
V = ml. sodium hydroxide used for sample.
v = ml. sodium hydroxide used for blank.
N = normality of sodium hydroxide.
W = sample weight represented by aliquot taken.

4.3.3 Electrical resistance.-The percussion-electric primers shall be tested for electrical resistance using an electrical circuit which limits the current through the primer to not more than 20 milliamperes with the electrode as one contact and the primer head as the other.

4.3.4 Non-functioning.

4.3.4.1 Percussion phase.-The primers shall be assembled into the test fixture (see 4.2.4). The fixture shall be set for a drop height of six inches and the load adjusted to give the required impact force.

4.3.4.2 Electrical phase.-The primers shall be tested using an electrical circuit which limits the current through the primer to 200 - 10 milliamperes with the plunger stem as one contact and the primer body as the other.

4.3.5 Functioning.

4.3.5.1 Percussion phase.-The test shall be performed as specified in 4.3.4.1 except that the fixture shall be set for a drop height of 20 inches.

4.3.5.2 Electrical phase.-The primers shall be subjected to the flow of an electrical current of 1.25 amperes maximum, using the plunger stem for one contact and the primer body as the other. Observation shall be made for functioning.

4.3.6 Firing.-The following test shall be performed at a Government proving ground.

4.3.6.1 Percussion.-The sample primers shall be assembled with a dummy projectile and percussion fired from a gun for which the round is standard.

4.3.6.2 Electrical.-The sample primers shall be assembled with a dummy projectile and electrically fired from a gun for which the round is standard.

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4.3.7 Determination of igniter charge composition.-The igniter charge composition, when butyl acetate is not used in the mixing process, shall be analyzed as follows to determine compliance with applicable drawing requirement.

4.3.7.1 Lead thiocyanate.-An accurately weighed portion of approximately 1 gm. of the sample shall be transferred to a tared medium - porosity sintered glass crucible. The contents of the crucible shall be extracted with five 20 ml. portions of 5 percent potassium nitrate solution, collecting the filtrate in a 500 ml. Erlenmeyer flask. The contents of the crucible shall be stirred after the addition of each portion and the potassium nitrate solution shall be allowed to remain in contact with the sample for a min. of 5 minutes before apply suction. The crucible and contents shall be washed with approximately 50 ml. of distilled water, dried in an oven at 100° to 105° C. for 2 hours, cooled in a desiccator and weighed. The above extraction shall be repeated with a 20 ml. portion of potassium nitrate solution and distilled water until the change in weight does not exceed 0.0005 gm. The crucible and contents shall be reserved for the lacquer determination (see 4.3.7.3). The loss in weight of the crucible and contents shall be calculated to percent of potassium chlorate plus lead thiocyanate in the sample on a moisture free basis. Exactly 35 ml. of standardized 0.1 normal (N) silver nitrate shall be added to the filtrate in the 500 ml. Erlenmeyer flask. The flask and contents shall be swirled and 100 ml. of 3N nitric acid (nitric acid used should be water-white and free of oxides of nitrogen) shall be added. The swirling of the mixture shall be continued for five minutes, 5 ml. of a saturated solution of ferric ammonium sulfate shall be added and the solution titrated with standardized 0.1N potassium sulphocyanate solution of the first permanent pale rose coloration. The percentage of lead thiocyanate in the sample shall be calculated on a dry basis as follows:

$$\text{Percent of lead thiocyanate} = \frac{35A-BC \times 16.17}{(W\ 1-M)}$$

where:

A = normality of the silver nitrate solution
B = volume in ml. of 0.1N potassium thiocyanate solution
C = normality of 0.1N potassium thiocyanate solution
W = weight of the sample
M = percent of moisture expressed as a decimal fraction

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4.3.7.2 Potassium chlorate.-The potassium chlorate content shall be determined by subtracting the percentage of lead thiocyanate from the percentage of combined lead thiocyanate plus potassium chlorate.

4.3.7.3 Lacquer.-The contents of the crucible reserved from the lead thiocyanate determination shall be extracted with five 20 ml. portions of warm acetone. The contents of the crucible shall be stirred after the addition of each portion and the acetone shall be allowed to remain in contact with the sample for a min. of two minutes before applying suction. The crucible and contents shall be aspirated until the odor of acetone is no longer perceptible and than dried in an oven at 100° to 105°C. for 1 hour, cooled in a desiccator and weighed. The above extraction shall be repeated with 20 ml. portions of acetone until the change in weight does not exceed 0.005 gm. The percentage of lacquer in the sample, on a dry basis, shall be calculated as follows:

$$\text{Percentage of lacquer} = \frac{(A-B) \times 100}{W\,(1-M)}$$

where:

A = weight of the crucible and residue before extraction with acetone
B = weight of the crucible and residue after extraction with acetone
M = percent moisture expressed as a decimal fraction
W = weight of the sample

4.3.7.4 Charcoal.-The percentage of charcoal in the sample shall be calculated on a dry basis as follows:

$$\text{Percentage of charcoal} = \frac{(A-B) \times 100}{W\,(1-M)}$$

where:

A = weight of crucible and residue after extraction with acetone
B = weight of empty crucible
M = percent moisture expressed as a decimal fraction
W = weight of the sample

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (see 6.1).
5.1.1 Level A.-Packaging for Primer, Percussion, Electric, M75, shall be in accordance with dwg. 8796678.
5.1.2 Level C.-Packaging for Primer, Percussion, Electric, M75, shall be in accordance with dwg. 8796535.
5.2 Packing (see 6.1).
5.2.1 Level A.-Packing for Primer, Percussion, Electric, M75, shall be in accordance with dwg. 8796679.
5.2.2 Level C.-Packing for Primer, Percussion, Electric, M75 shall be in accordance with dwg. 8796536.
5.3 Marking.
5.3.1 Level A.-Marking for interior packages shall be in accordance with dwg. 8796678. Marking of exterior boxes shall be in accordance with dwg. 8796679.
5.3.2 Level C.-Marking of interior packages is not required. Marking of exterior boxes shall be in accordance with dwg. 8796536.
5.4 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:
a. Title, number, and date of this specification.
b. Level of protection required.
6.2 Check test expense.-The contracting officer will arrange to have the contractor reimbursed for the expense incurred in the performance of the check test.
6.3 Check test for possible deterioration of percussion artillery primers.-The following tests are designed to prevent the loading of artillery primers which have deteriorated into a major ammunition item. The test shall be performed on artillery primers from each lot which has been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the artillery primers are loaded does not require that the check test be performed, the contracting officer should insert a provision for the performance of the check tests in the contract.

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6.3.1 Non-functioning percussion phase.-The number of samples, acceptance and rejection shall be as specified under the sampling plan designated by the code letter H, AQL of 0.40 percent of Standard MIL-STD-105. The samples shall be tested as specified in 4.3.4.1.

6.3.2 Non-functioning electrical phase.-The sampling plan for this test shall be as specified in 6.3.1. The test shall be performed as specified in 4.3.4.2.

6.3.3 Functioning percussion phase.-The number of samples, acceptance and rejection shall be as specified under the sampling plan designated by the code letter O with AQL of 0.40 percent of Standard MIL-STD-105. The samples shall be tested as specified in 4.3.5.1.

6.3.4 Functioning electrical phase.-The sampling plan for this test shall be as specified in 6.3.3. The test shall be performed as specified in 4.3.5.2.

6.3.5. Electrical resistance.-The contractor shall perform this test on all the samples selected for and prior to, the tests specified in 6.3.1, 6.3.2, 6.3.3 and 6.3.4. If any of these primers fail the requirement of 3.5, the entire lot of primers shall be tested prior to use in a complete round lot of ammunition.

6.3.6 Moisture content of black powder.-A sample of 25 primers shall be selected for this test. The black powder shall be removed from the flash tube and the moisture content determined* in accordance with the method specified in Specification JAN-P-223. If the moisture content of any of the powder tested exceeds 0.7 percent, the lot of primers shall be rejected.

*Since black powder is hygroscopic, it is advisable to place the powder in a dry air-tight container immediately upon removal, and to run the determination as soon as possible thereafter.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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